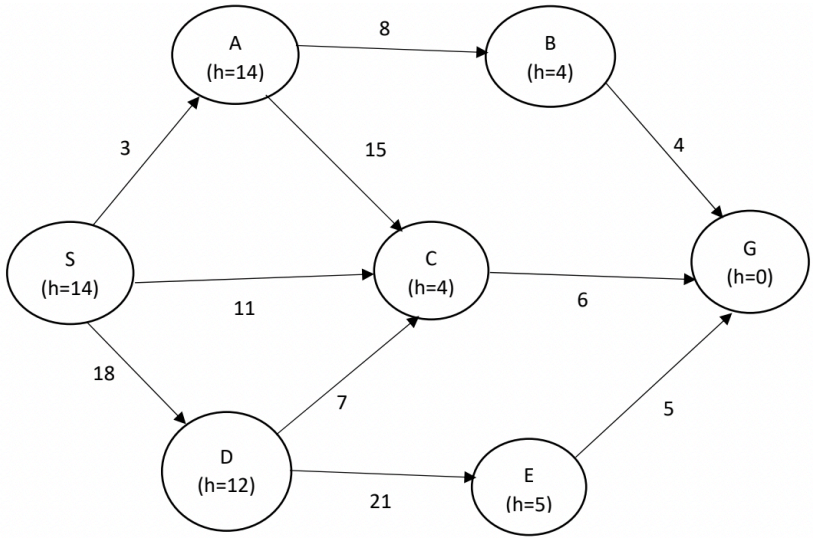
1. (a)Consider the following state-space graph:



Here S is the starting node and G is the goal node. Now change only the heuristic values of any two nodes so that the heuristic values become both admissible and consistent. Mention the updated values and corresponding node names. Explain your answer. [4]

(b) Find out the solution paths and costs returned by the following search algorithms: i) Greedy Best First Search ii) A* Tree Search iii) A* Graph Search [3]

(c) Suppose you have three heuristic functions h1, h2 and h3. h1 and h2 are admissible but h3 is inadmissible. Consider the following heuristics:
h4(n)=max(h1, h2, h3)
h5(n)=min(h1, h2, h3)
h6(n)=(h1+h2)/2
i. Which heuristic is inadmissible? Why? [1.5]
ii. Propose a heuristic that will be better than h4, h5 and h6. [1.5]

Local Search

Courier Chris runs a package delivery service that covers 10 major cities. Chris needs to plan a **round trip**—starting from one city, visiting all other cities **exactly once**, and returning to the starting city.

Chris wants to **Minimize** total distance traveled (fuel costs) and to **Minimize** overall driving time (improving delivery speed). The **goal** is to find the **shortest** or most efficient route that visits all 10 cities.

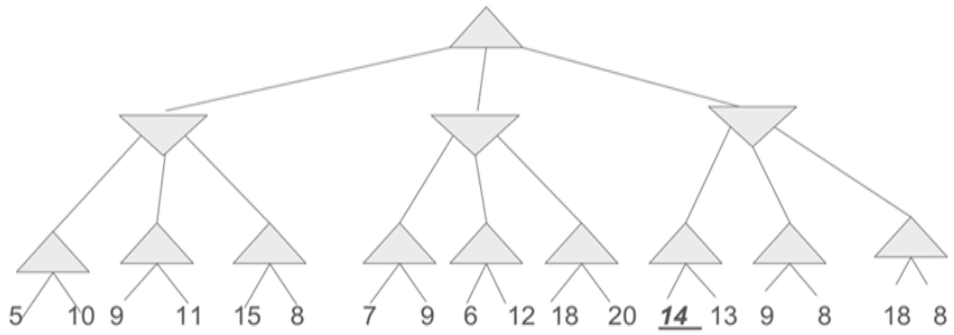
a. Construct the cost function to solve the problem using the simulated annealing algorithm? [1]

b. What changes you may need to make on the cost function you constructed, if you were to solve the problem using usual Hill Climbing algorithm. [2]
c. Use this scenario to explain the concept of “Plateau” in Hill Climbing. [3]
d. How does simulated annealing allow Courier Chris to potentially escape from the plateau? In your answer, define the concept of “temperature” and describe why, at higher temperatures, even worse solutions might be temporarily accepted. [4]

Set A (Game playing)

a. Using alpha beta pruning technique, find out the branches of the following game tree which can be avoided from search to find the best move 7

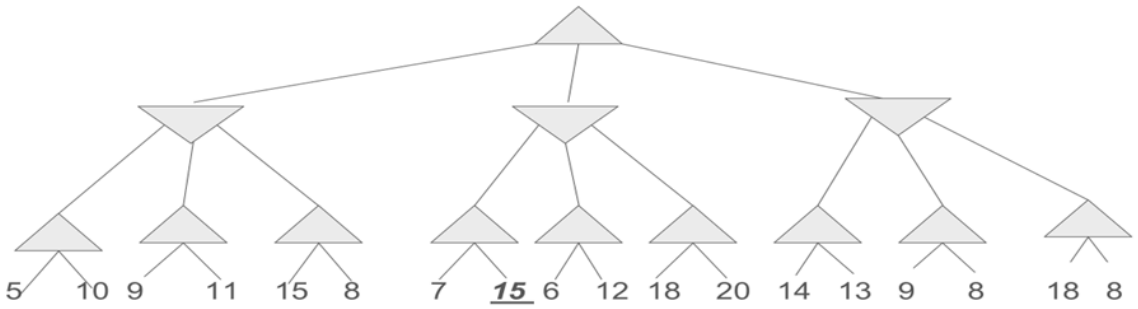
b. Replace the underlined utility value shown in the following game tree so that the rightmost two branches are avoided from search. 3



Set B (Game playing)

a. Using alpha beta pruning technique, find out the branches of the following game tree which can be avoided from search to find the best move 7

b. Replace the underlined utility value shown in the following game tree so that the two branches right to it are avoided from search. 3



Genetic Algorithm

x_1	x_2	y
1	2	2
2	4	4
3	6	6

Suppose you are given a regression problem, where for different values of X_1 and X_2 you have definite values of Y. You are asked to calculate the optimal values of the regression coefficients to get the best fit straight line, but you will do this by applying **Genetic Algorithm.**

Hints: Equation of a straight line in this case will be **$y = m_1x_1 + m_2x_2 + c$, so m_1, m_2 and c are the regression coefficients.** You will use the sum of the squared residuals $\sum(y - y')^2$ as the loss function. The best fit straight line is the line that minimizes the values of the loss function the most.

a. Encode the problem to create 3 chromosomes. [3]
b. Apply an appropriate fitness function and calculate the fitness of the chromosomes derived in (a). Additionally select the best two chromosomes [3]

c. Using the results from (b), apply single point crossover to get 2 children. [2]

d. Perform mutation to get the final pair of children, calculate the final fitness and comment on the fitness of the newly produced offspring. [2]